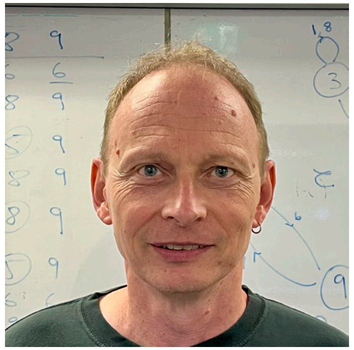
Artificial Intelligence

Introduction and (brief) history of AI

Lecturers

- Dr Francisco Cruz (LiC – Lecturer in Charge)
  - https://www.unsw.edu.au/staff/francisco-cruz-naranjo

- A/Prof Oliver Obst (Co-Lecturer)
  - https://www.unsw.edu.au/staff/oliver-obst



Admin Team

- Maher Mesto (Course admin)
  - m.mesto@unsw.edu.au

- John Chen (Forum monitoring)
  - xin.chen9@student.unsw.edu.au

Tutors

• Adam Stucci	a.stucci@unsw.edu.au
• Chunyang Meng	chunyang.meng@unsw.edu.au
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• Joffrey Ji	joffrey.ji@unsw.edu.au
• Peter Ho	peter.ho2@student.unsw.edu.au
• Oltan Sevinc	m.sevinc@unsw.edu.au
• Chris Lee	christopher.lee1956@student.unsw.edu.au
• Markus Ernst	m.ernst@unsw.edu.au
• Yi Luo	yi.luo7@student.unsw.edu.au

- **Session will be BYOD. Alternatively, you can borrow a laptop. See https://taggi.cse.unsw.edu.au/FAQ/Borrow_A_Laptop/**

Course Plan

Week	Lecture	Tutorial
1	Introduction/Agents/Search	
2	Planning	Search algorithms
3	Knowledge representation	Planning
4	Neural networks	Knowledge representation and reasoning
5	Reinforcement learning	Biomass growing
6	Flexibility week	A1 discussion
7	Reasoning with uncertainty	TD learning
8	Computer vision	Bayesian nets
9	Foundation models	Open CV
10	Human-aligned intelligent robotics	Foundation models / A2 discussion

Related Course

- COMP3431 Robot Software Architectures
- COMP4418 Knowledge Representation and Reasoning
- COMP9417 Machine Learning and Data Mining
- COMP9444 Neural Networks and Deep Learning
- COMP9491 Applied Artificial Intelligence
- COMP9517 Computer Vision
- COMP6713 Natural Language Processing
- COMP9418 Advanced Machine Learning
- COMP9727 Recommender Systems

Timetable

- Lecture:
  - Fridays
  - From 18h to 21h
- Tutorials:

Class Code	Section	Location	Time	Tutor	Email
4421	M09A	Online (Blackboard Collaborate)	Mon 09:00 - 11:00 (Weeks: 2-10)	Oltan Sevinc	m.sevinc@unsw.edu.au
4422	M11A	Quadrangle G052 (K-E15-G052) *	Mon 11:00 - 13:00 (Weeks: 3-10) *	Joffrey Ji	joffrey.ji@unsw.edu.au
4423	W09A	Newton 307 (K-J12-307)	Wed 09:00 - 11:00 (Weeks: 2-10)	Peter Ho	peter.ho2@unsw.edu.au
4425	W11A	Tyree Energy Technology LG05 (K-H6-LG05)	Wed 11:00 - 13:00 (Weeks: 2-10)	Hadha Afrisal	hadha.afrisal@unsw.edu.au
4426	W11B	Mathews 226 (K-F23-226)	Wed 11:00 - 13:00 (Weeks: 2-10)	Haitao Gao	haitao.gao@unsw.edu.au
4428	W13A	Mathews 309 (K-F23-309)	Wed 13:00 - 15:00 (Weeks: 2-10)	John Chen	xin.chen9@student.unsw.edu.au
4429	W13B	Old Main Building 151 (K-K15-151)	Wed 13:00 - 15:00 (Weeks: 2-10)	Chris Lee	christopher.lee1956@student.unsw.edu.au
4430	W13C	Mathews 226 (K-F23-226)	Wed 13:00 - 15:00 (Weeks: 2-10)	Markus Ernst	m.ernst@unsw.edu.au
4431	W18A	Quadrangle 1001 (K-E15-1001)	Wed 18:00 - 20:00 (Weeks: 2-10)	Yi Luo	yi.luo7@student.unsw.edu.au
4432	W18B	Online (Blackboard Collaborate)	Wed 18:00 - 20:00 (Weeks: 2-10)	John Chen	xin.chen9@student.unsw.edu.au
4419	H18A	Mathews 306 (K-F23-306)	Thu 18:00 - 20:00 (Weeks: 2-10)	Joffrey Ji	joffrey.ji@unsw.edu.au
4415	F11A	AGSM LG06 (K-G27-LG06)	Fri 11:00 - 13:00 (Weeks: 2-10)	Adam Stucci	a.stucci@unsw.edu.au
4416	F11B	Morven Brown LG2 (K-C20-LG2)	Fri 11:00 - 13:00 (Weeks: 2-10)	Khaja Ahmed Shaik	k.shaik@unsw.edu.au
4417	F13A	AGSM LG06 (K-G27-LG06)	Fri 13:00 - 15:00 (Weeks: 2-10)	Chunyang Meng	chunyang.meng@unsw.edu.au
4418	F13B	Webster 302 (K-G14-302)	Fri 13:00 - 15:00 (Weeks: 2-10)	Stephen Xie	xiongyu.xie@unsw.edu.au

- See updated version at: https://moodle.telt.unsw.edu.au/mod/page/view.php?id=8704573

Important dates

- **First lecture:** Friday 5^{th} June 2026
- **Flexibility week:** Friday 10^{th} July 2026
- **Last lecture:** Friday 7^{th} August 2026
- **Assignment 1:** open by Week 2, deadline Week 5 (discussions in week 6)
- **Assignment 2:** open by Week 6, deadline Week 9 (discussions in week 10)
- **Exam:** Exam period

Assessment

- Assessment will consist of:
  - Assignment 1: 21%.
  - Assignment 2: 20%.
  - Tutorial work: 9%.
  - Final exam 50%.
    - In-person, MCQs, BYOD using Inspera, with 10% discount for incorrect answers.
- To pass, you must score:
  - A combined mark of at least 50/100.
  - At least 20/50 for the exam (or 40%).

Student Conduct

- Assignments will be done individually.
  - Students must participate in the discussion.
- Late deliveries will be accepted subject to 5% discount per day from the results (including weekends and public holidays), for up to 5 days, after which mark is 0.
- It’s students' responsibility to have code discussions with tutors in time.
- Plagiarism is academic misconduct.

Contact

- The first contact should be the forums.
- In special circumstances, you could also email:
  - Your tutor
  - The course admin (m.mesto@unsw.edu.au)
  - The co-lecturer (o.obst@unsw.edu.au)
  - The lecturer in charge (f.cruz@unsw.edu.au)

Texts & References

- Poole, D.L. & Mackworth, A. Artificial Intelligence: Foundations of Computational Agents. Second Edition. Cambridge University Press, Cambridge, 2017.
- Russell, S.J. & Norvig, P. Artificial Intelligence: A Modern Approach. Fourth Edition, Pearson Education, Hoboken, NJ, 2021.
- Sutton, R. & Barto, A. Reinforcement Learning: An Introduction. MIT press. 2018.
- Jurafsky, D. & Martin, J. H. Speech and Language Processing. Stanford. 2023.

Artificial Intelligence

A (brief) history of AI

UNSW SYDNEY

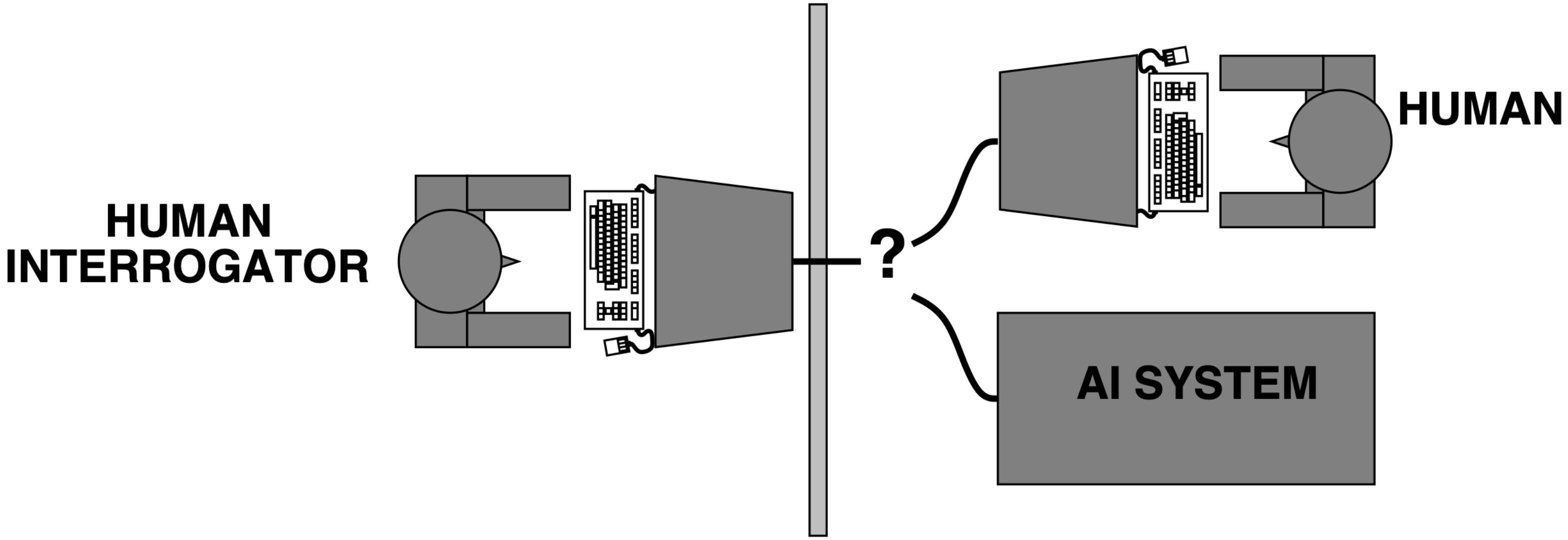
Artificial Intelligence: The First 2,400 years

• Logic (Aristotle c. 350BC, Boole 1848, Frege 1879, Tarski 1935)	Deductive reasoning, problem solving
• Formal algorithms (Euclid c. 300BC)	Theoretical foundations
• Inductive Reasoning (Bacon 16th C, Hume 18th C)	Machine Learning
• Probability (Pascal 17th C, Bayes 18th C)	Uncertain reasoning
• Utility theory (Mill 1863)	Uncertain reasoning, reinforcement learning
• Structural linguistics (Saussure 1916, Bloomfield 1933)	Natural Language Processing
• Formal systems (Gödel 1929, Turing 1936)	Theoretical foundations
• Neural networks (McCullogh & Pitts 1943)	Machine Learning
• Cybernetics/Control theory (Wiener 1948)	Autonomous systems
• Game theory (von Neumann & Morgernstern 1947)	Game play, adversarial agents
• Decision theory (Bellman 1957)	Reinforcement learning
• Formal linguistics (Chomsky 1957)	Natural Language Processing

Milestones

- 1950 – Turing's paper on the imitation game and the child machine
- 1956 – Dartmouth Conference (first use of "Artificial Intelligence")
- 1960 – Adaptive Control (Widrow & Hoff)
- 1962 – Perceptron (Rosenblatt)
- 1963 – Symbolic Algebra (Slagle)
- 1969 – Shakey (Nilsson, Duda, Hart, Fikes, … SRI International)
- 1975 – Freddy and the AI Winter (Michie *et al*, Edinburgh)
- 1981 – First International Workshop on Machine Learning
- 1981 – Decision Trees (Quinlan, Stanford, USyd, NSWIT, UNSW)
- 1997 – Deep Blue beats Gary Kasparov (IBM)
- 1997 – First RoboCup competition
- 2018 – OpenAI's GPT-1 released

Turing's Imitation Game (1950)



I believe that in about fifty years time [2000] it will be possible to programme computers to make them play the imitation game so well that an average interrogator will not have more than 70 per cent chance of making the right identification after five minutes of questioning.

Eliza Session

Doctor: What brought you here to see me today?
Visitor: Oh, nothing much. I’m feeling a bit tired, that’s all.

Doctor: Why do you think you’re feeling a bit tired?
Visitor: Well, I’ve been traveling a lot, and away from home.

Doctor: Tell me more about your family.

Eliza Session

Visitor: The trouble is, my mother’s ill.

Doctor: How long has she been ill?

...

Visitor: The trouble is, my mother’s Irish.

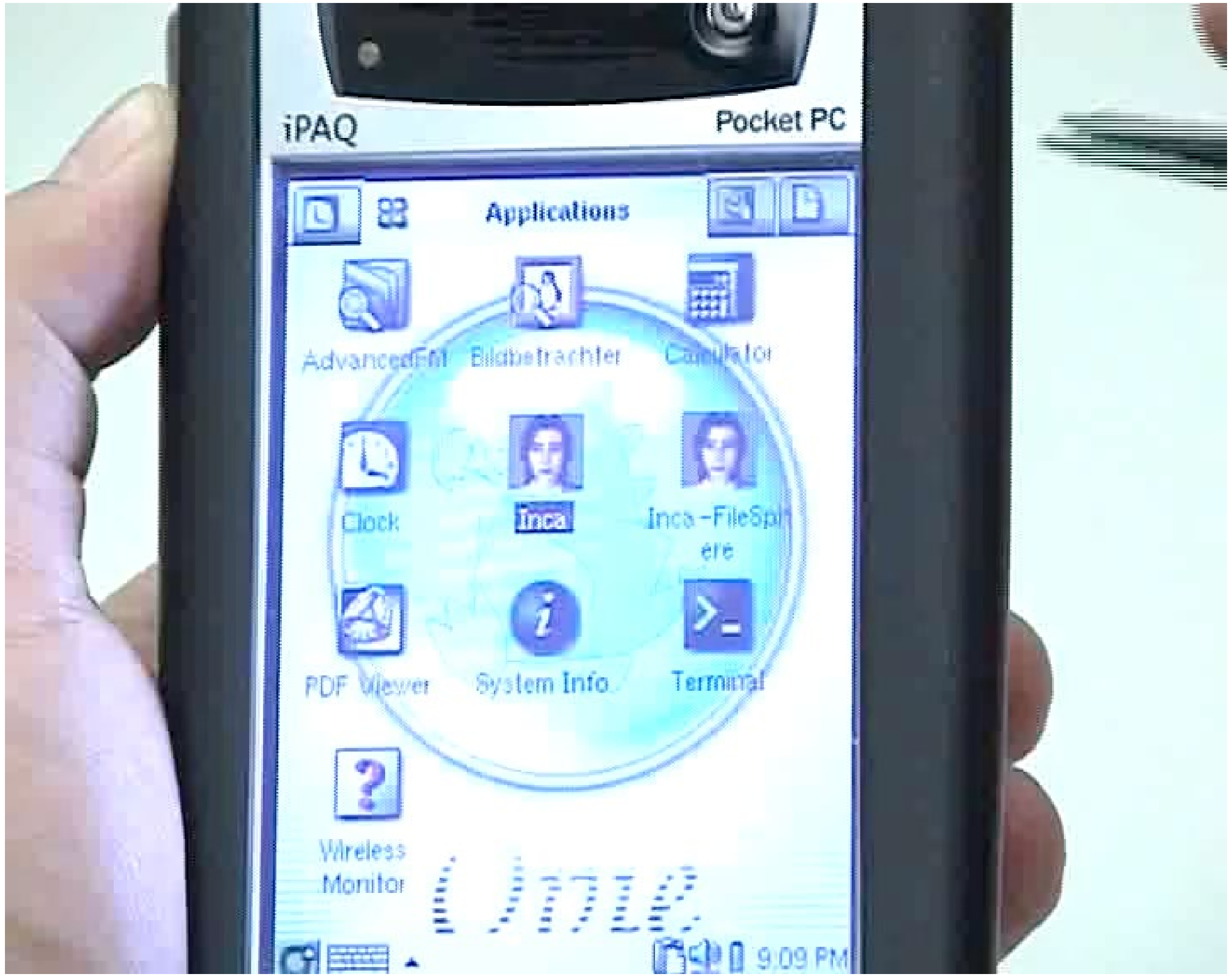
Doctor: How long has she been Irish?

...

Visitor: The trouble is, my mother’s poodle.

Doctor: How long has she been poodle?

Conversational Agents



The Child Machine

“Instead of trying to produce a programme to simulate the adult mind, why not rather try to produce one which simulates the child's? If this were then subjected to an appropriate course of education one would obtain the adult brain. Presumably the child-brain is something like a notebook as one buys from the stationers. Rather little mechanism, and lots of blank sheets... Our hope is that there is so little mechanism in the child-brain that something like it can be easily programmed. The amount of work in the education we can assume, as a first approximation, to be much the same as for the human child.”

Alan Turing (1950)
Computing Machinery and Intelligence

Agents and Autonomous Systems

- Complex behaviours in dynamic environments
- Have to integrate almost all aspects AI
- Combines computing with many other disciplines

Autonomous Systems

Shakey - The First Integrated AI System

Artificial Intelligence (AI)

- What is intelligence?

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  - it can be described as the ability to **perceive** or infer **information**, and to retain it as knowledge to be applied towards adaptive behaviours within an environment or context [Wikipedia].

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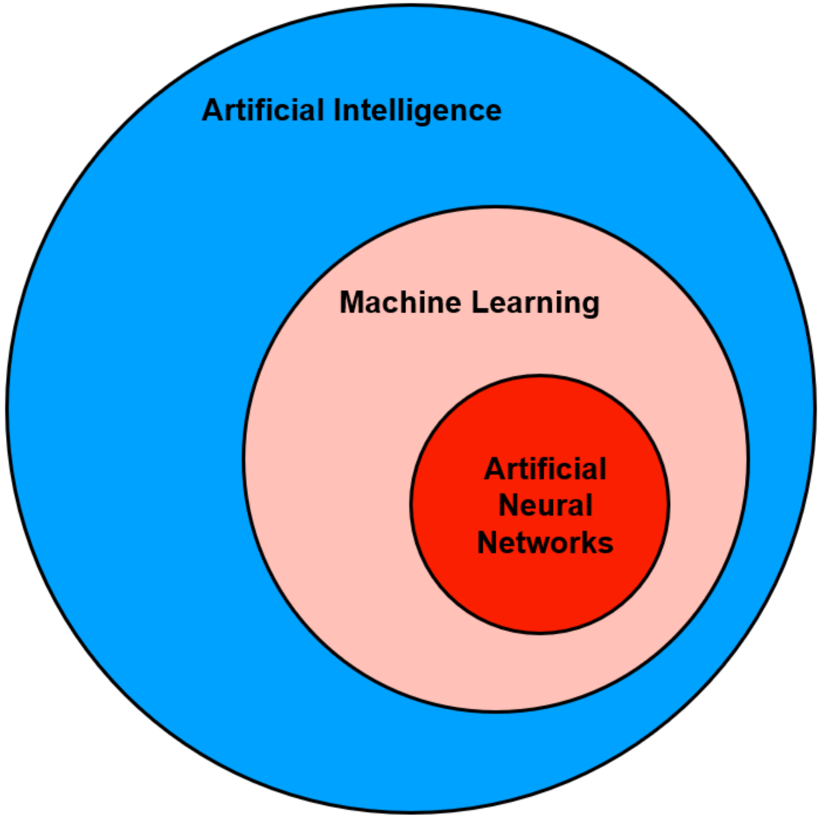
- What is artificial intelligence?
  - Artificial intelligence (AI) is intelligence demonstrated by machines, as opposed to intelligence displayed by humans or by other animals [Wikipedia].

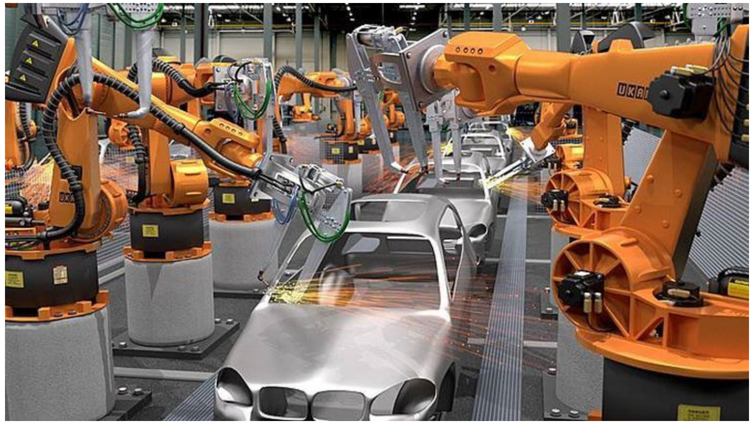
AI is not ML is not ANN

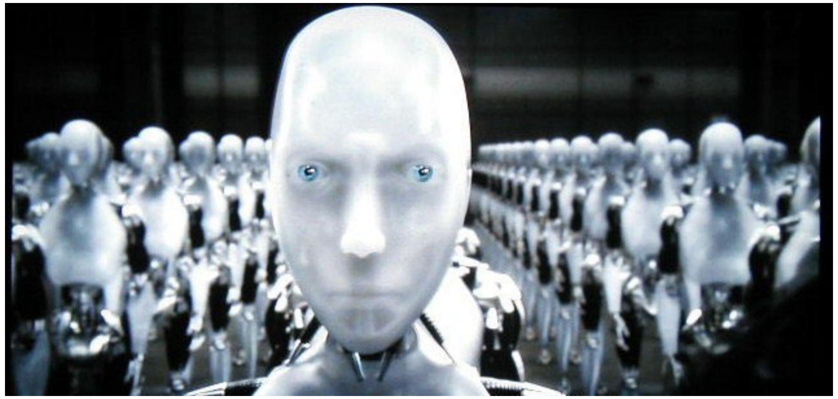


AI is not Python

Current (mis)understanding of AI

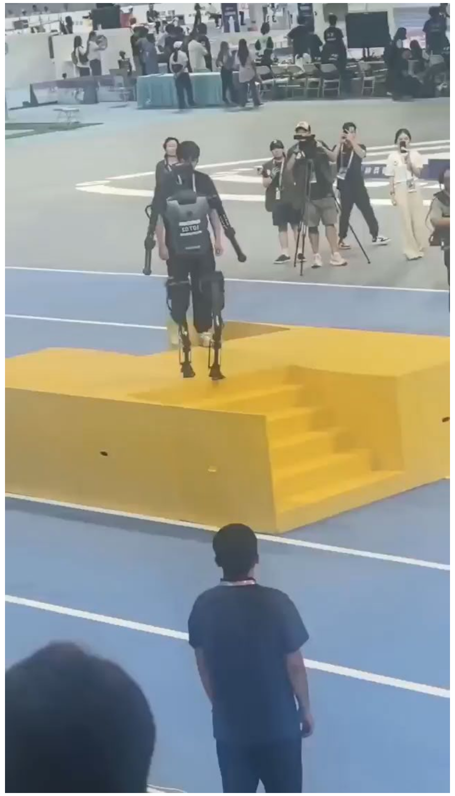
How are AI-based systems currently perceived?





Current (mis)understanding of AI

But in reality, it's still an open problem



Current (mis)understanding of AI

But in reality, it's still an open problem…

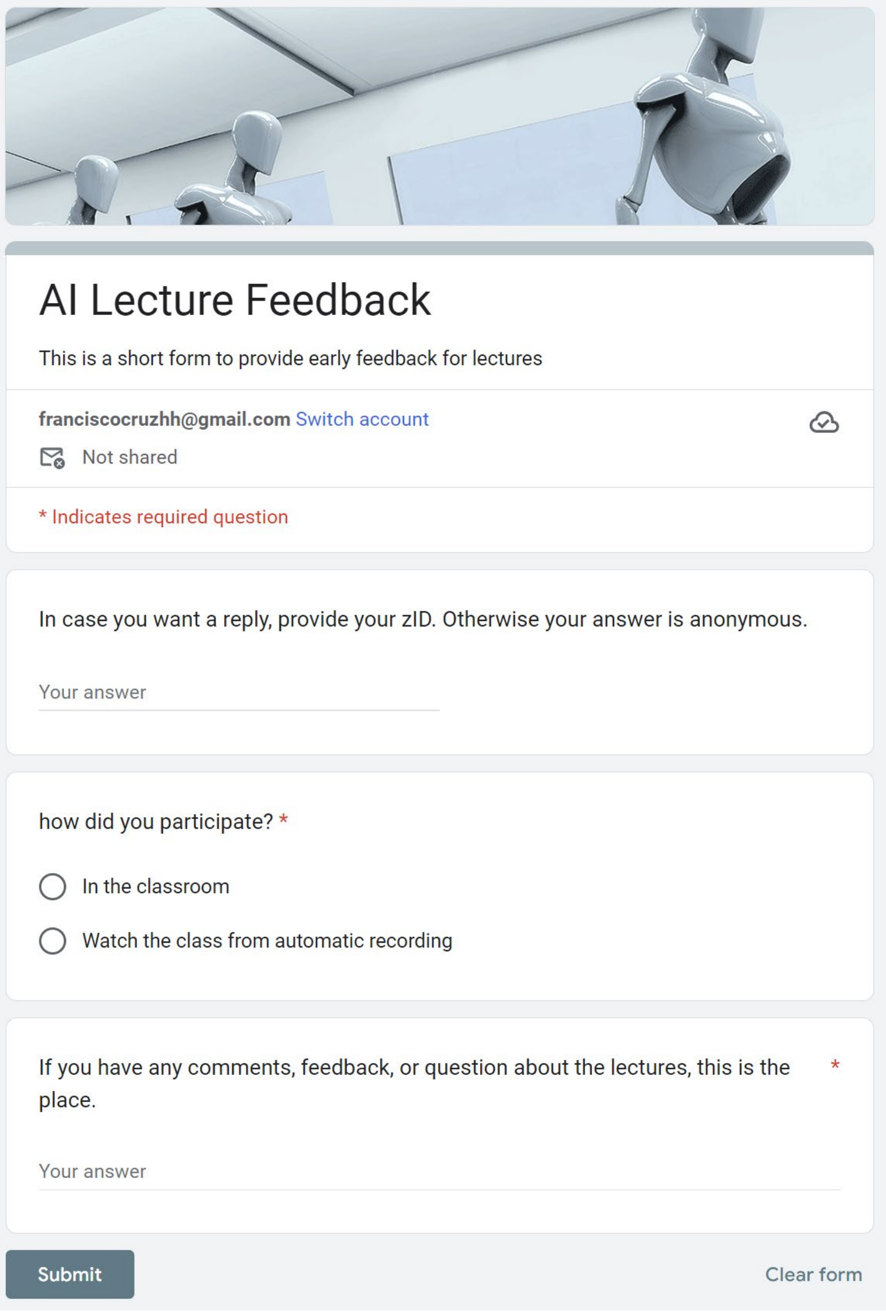
Beyond dancing robots!

Recent History

- Read Nils Nilsson's "The Quest for Artificial Intelligence"
- https://ai.stanford.edu/~nilsson/QAI/qai.pdf

Feedback

- In case you want to provide anonymous feedback on these lectures, please visit:
- https://forms.gle/KBkN744QuffuAZLF8


AI Lecture Feedback

This is a short form to provide early feedback for lectures

franciscocruzhh@gmail.com Switch account

Not shared

* Indicates required question

In case you want a reply, provide your zID. Otherwise your answer is anonymous.

Your answer

how did you participate? *

- In the classroom
- Watch the class from automatic recording

If you have any comments, feedback, or question about the lectures, this is the place. *

Your answer

Submit

Clear form